

Deep Learning for Pelvic Organ Segmentation in Radiotherapy: A Systematic Review

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ABSTRACT

Background and Purpose: Accurate organ segmentation is a fundamental component of radiotherapy planning, particularly in the pelvic region where target volumes and multiple organs-at-risk must be delineated with high precision. This systematic review synthesizes evidence from 173 peer-reviewed studies to characterize recent developments in deep learning methods and to assess the methodological reliability of the evidence supporting their clinical adoption.

Methods: The review systematically examines architectural trends, organ-specific performance, dataset characteristics, and evaluation practices. Methodological reliability was assessed using a structured framework based on indicators related to validation strategies, reproducibility, and clinical relevance.

Results: Convolutional encoder-decoder architectures, predominantly U-Net and its variants, continue to dominate the literature, frequently incorporating residual connections, attention mechanisms, or hybrid CNN-Transformer components. Segmentation accuracy remains highly structure-dependent: while large organs with well-defined boundaries (e.g., bladder, femoral heads) consistently achieve high scores in overlap-based metrics such as the Dice Similarity Coefficient (DSC), smaller or poorly contrasted structures (e.g., penile bulb, seminal vesicles) present persistent challenges. Despite the prevalence of CT imaging, MRI and cone-beam CT (CBCT) are increasingly utilized to enhance soft-tissue definition and support adaptive workflows.

Conclusions: The analysis reveals a significant gap in model generalizability. Although independent test-set evaluation is common (94%), external validation across independent institutions and multi-center dataset utilization remain limited. Furthermore, reproducibility is hampered by low rates of code and dataset sharing, and clinical or dosimetric evaluations are not yet routinely integrated. While technical progress is substantial, further advancement depends on improving dataset diversity, methodological transparency, and clinically oriented evaluation frameworks to facilitate the transition of automated segmentation into a reliable tool for routine radiotherapy.

1. Introduction

Radiotherapy is a fundamental modality in cancer treatment, relying on the precise delivery of ionizing radiation to eradicate tumor cells while preserving surrounding healthy tissues. The treatment planning process is guided by medical imaging, primarily computed tomography (CT) and magnetic resonance imaging (MRI), which enable the identification of target volumes and organs at risk (OAR) [1, 2]. Modern radiotherapy techniques allow highly conformal dose distributions, increasing the dependence on accurate anatomical delineation. In this context, segmentation and contouring are essential steps for ensuring treatment effectiveness while minimizing radiation-induced toxicity [3].

Although often used interchangeably, contouring and segmentation refer to related but distinct processes. Contouring typically denotes the manual delineation of anatomical structures by clinicians, including gross tumor volume (GTV) and OARs, whereas segmentation encompasses automated or semi-automated methods that partition medical images into meaningful regions [4, 5, 6]. The accuracy of these processes directly affects treatment outcomes, as imprecise boundaries may result in suboptimal dose distributions or unintended irradiation of healthy tissues. These challenges are particularly evident in the pelvic region, where multiple critical structures are located in close proximity and imaging contrast is frequently limited.

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Pelvic cancers, including prostate, cervical, rectal, and bladder malignancies, represent a major clinical domain for radiotherapy [7]. In this anatomical region, manual contouring is time-consuming and subject to significant inter-observer variability, while anatomical variability and organ motion further complicate the delineation process. Multimodal imaging strategies, such as the integration of MRI with CT, can improve soft-tissue visualization but also increase the complexity of clinical workflows. Consequently, reliable segmentation methods are essential to support treatment planning and to reduce treatment-related toxicities, including urinary, gastrointestinal, and sexual dysfunctions [8, 9].

To address these challenges, machine learning (ML) and deep learning (DL) techniques have been increasingly adopted for automated segmentation. Early ML approaches relied on handcrafted features and showed limited generalization across imaging modalities [6, 10, 11]. In contrast, deep learning models, particularly convolutional neural networks (CNNs) such as U-Net and its variants, have demonstrated strong performance in medical image segmentation by learning hierarchical representations directly from imaging data [12, 13, 14]. Despite these advances, important challenges remain, including the need for large annotated datasets, high computational requirements, and variability in model performance across clinical settings [15, 16, 17].

An additional aspect that has received comparatively limited attention is the role of model optimization. Hyperparameters such as learning rate, batch size, and architectural configuration strongly influence training dynamics, convergence, and generalization. Inadequate tuning can lead to suboptimal performance, particularly in anatomically complex regions such as the pelvis. Although hyperparameter optimization has been extensively studied in the broader machine learning literature [18, 19, 20], its systematic application and reporting in radiotherapy segmentation studies remain limited.

Several systematic reviews have examined deep learning approaches for organ segmentation in radiotherapy, reporting promising results in tasks such as clinical target volume delineation in prostate cancer [21, 22]. However, most of these reviews adopt a generalized anatomical perspective or focus primarily on performance metrics, with limited discussion of pelvic-specific challenges such as anatomical variability, organ motion, and inter-observer variability. Furthermore, the role of optimization strategies and hyperparameter tuning in deep learning-based pelvic organ segmentation has received little attention in previous reviews.

These limitations highlight the need for a more focused and integrative analysis of recent advances in deep learning-based segmentation for pelvic radiotherapy. The objective of this systematic review is to synthesize current developments in deep learning methods for pelvic organ segmentation, with particular attention to architectural design, organ-specific segmentation behaviour, dataset characteristics, optimization practices, evaluation frameworks, and methodological aspects that influence reproducibility and clinical translation. By jointly examining these dimensions, this study aims to provide a clearer understanding not only of how high segmentation accuracy has been achieved, but also of the factors that continue to limit the reliability, generalizability, and clinical adoption of automated segmentation systems in pelvic radiotherapy.

To the best of our knowledge, no previous systematic review has jointly analyzed deep learning architectures, optimization strategies, dataset characteristics, and evaluation practices in the context of pelvic radiotherapy segmentation.

To guide this analysis, the following research questions were formulated to examine both the technical characteristics of deep learning approaches and the methodological reliability of the evidence reported in the literature:

- **RQ1:** What deep learning architectures are most commonly used for pelvic organ segmentation in radiotherapy, and which design strategies are associated with improved segmentation performance?
- **RQ2:** How does segmentation performance vary across pelvic anatomical structures, and what organ-specific factors influence model difficulty and clinical reliability?
- **RQ3:** How do dataset characteristics, including imaging modality, dataset scale, and data diversity, influence model performance and generalizability in pelvic organ segmentation?
- **RQ4:** How are optimization strategies and hyperparameter tuning practices applied and reported in deep learning models for pelvic organ segmentation?
- **RQ5:** What evaluation practices are used to assess segmentation quality, and to what extent do they capture clinically meaningful outcomes?
- **RQ6:** What methodological limitations, evidential reliability issues, and research challenges emerge from the current literature on deep learning-based pelvic organ segmentation?

The remainder of this paper is organized as follows. Section 2 describes the methodology of the systematic review. Section 3 presents the results and synthesis of the selected studies. Section 4 discusses the findings through an integrated thematic analysis informed by the research questions. Finally, Section 5 presents the conclusions and outlines key directions for future research.

2. Materials and Methods

2.1. Methodology

This systematic literature review was conducted following the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [23], which provide a structured framework to ensure transparency and methodological rigor in evidence synthesis. The review process comprised four main stages: literature search, study screening, eligibility assessment, and structured data extraction and analysis.

A comprehensive literature search was performed in Scopus and Web of Science (WoS) up to December 2024. These databases were selected due to their broad coverage of both medical imaging and computational research. The search strategy was designed to capture studies addressing deep learning-based organ segmentation in pelvic radiotherapy. The search queries combined terms related to radiotherapy, pelvic anatomical structures, organ segmentation, and deep learning techniques. The complete search criteria and keyword combinations are summarized in Table 1.

The initial screening stage consisted of reviewing titles and abstracts to identify studies relevant to the objectives of the review. Two independent reviewers performed this screening process. Discrepancies were resolved through discussion and, when necessary, consultation with a third reviewer.

Studies that met the inclusion criteria were subsequently assessed through full-text review. Relevant information was extracted using a structured data extraction template designed to capture key methodological and technical aspects of each study. Extracted variables included study objectives, segmentation architectures, dataset characteristics, imaging modalities, optimization strategies, and evaluation metrics. Only peer-reviewed articles written in English and indexed in Scopus or Web of Science were included in the final dataset.

The extracted information was analyzed using a predefined analytical framework aligned with the research questions of the review. This framework comprised two complementary groups of analytical dimensions. The first group focused on the technical and methodological characteristics of the proposed segmentation approaches, including model architectures, cancer types addressed, imaging modalities, evaluation metrics, optimization procedures, and dataset characteristics. These variables form the basis of the descriptive synthesis presented in the Results section.

The second group consisted of methodological indicators related to evidential reliability, including validation strategies, dataset diversity, annotation practices, reproducibility, and clinical evaluation. These indicators were used to assess the methodological robustness of the reviewed studies and to characterize the evidential maturity of the field.

Based on these criteria, each study was assigned to one of three overall evidential strength categories (higher, moderate, or lower evidential strength). Studies classified as higher evidential strength satisfied most reliability indicators, including external validation and transparent reporting practices. Moderate evidential strength corresponded to studies meeting several methodological criteria but lacking elements such as external validation or reproducibility resources. Lower evidential strength referred to studies with limited reporting of validation procedures, annotation practices, or experimental transparency.

Data processing and aggregation were supported by custom scripts implemented in Python to ensure consistency and traceability throughout the analysis.

2.2. Search Strategy

The search strategy was designed to ensure comprehensive coverage of studies on deep learning-based organ segmentation in pelvic radiotherapy. Keywords related to radiotherapy, organ segmentation, machine learning, and anatomical focus were combined using Boolean operators (AND, OR) and applied to titles, abstracts, and keywords, as detailed in Table 1. The selection process is summarized in Figure 1.

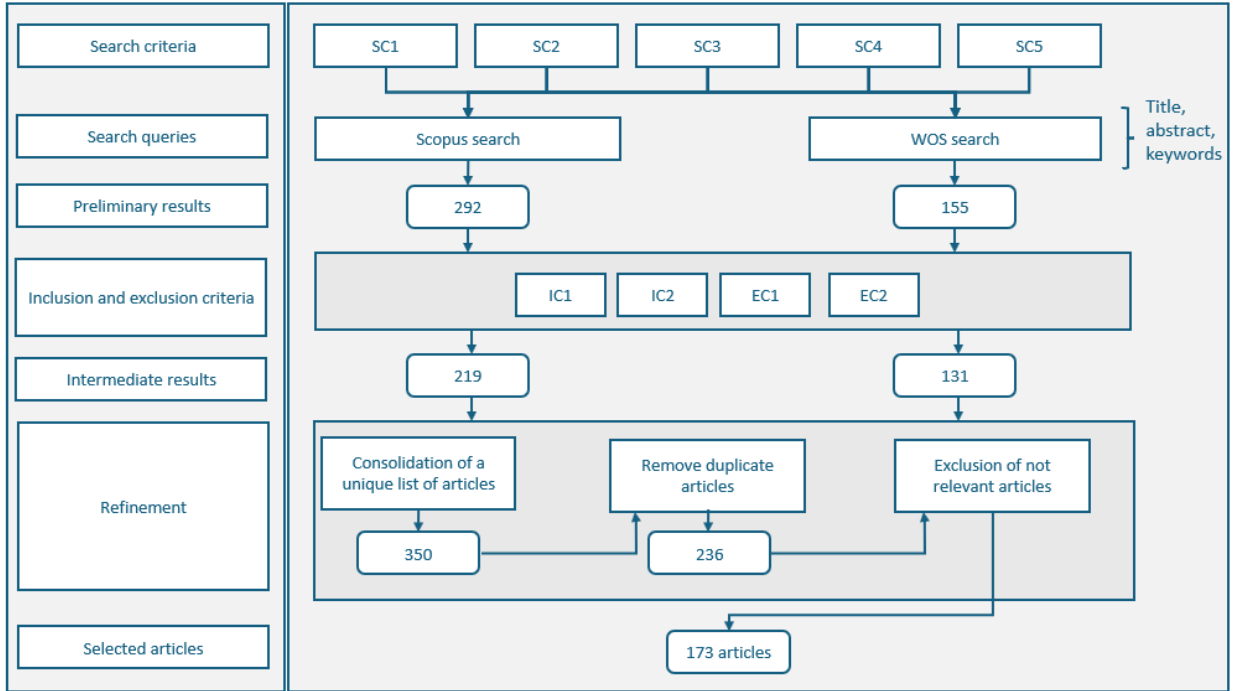


Figure 1: Workflow for the retrieval of relevant articles.

The primary search was conducted in Scopus and Web of Science (WoS), which were selected because they provide broad multidisciplinary coverage of peer-reviewed journals across both biomedical research and computational sciences. Recent large-scale comparisons of bibliographic data sources have confirmed that Scopus and Web of Science remain among the most comprehensive and widely used databases for indexing scholarly literature across multiple scientific disciplines [24, 25]. In addition, methodological studies evaluating academic search systems for evidence synthesis have identified these databases as suitable platforms for systematic searches due to their structured indexing, support for advanced query formulation, and reproducibility of search strategies [26]. Scopus and Web of Science were therefore used as the main platforms for identifying relevant studies within the predefined search strategy. A large proportion of journals indexed in PubMed and IEEE Xplore are also covered by Scopus and Web of Science, resulting in substantial cross-database coverage of peer-reviewed literature in medical imaging and artificial intelligence. Consequently, studies indexed in PubMed or IEEE Xplore were retrieved indirectly through their inclusion in Scopus or Web of Science during the search and screening process. No studies were manually added from those databases outside the predefined search protocol. To further ensure transparency and completeness of the search strategy, an additional verification step was conducted. The journals in which the included studies were published were checked to confirm their indexing in Scopus and/or Web of Science at the time of the search. In addition, exploratory searches using the same core keywords were performed in PubMed and IEEE Xplore to identify potentially relevant records uniquely indexed in those databases. These exploratory checks did not identify additional studies meeting the predefined inclusion criteria of this review.

The initial search yielded **292 articles from Scopus** and **155 from Web of Science**, resulting in **447 records**. Inclusion and exclusion criteria (Table 2) were then applied.

After applying these criteria, **219 articles from Scopus** and **131 from WoS** were retained, resulting in **350 records**. The retrieved studies were consolidated into a structured dataset including metadata such as title, authors, abstract, keywords, DOI, publication year, and journal information.

Duplicate records were removed using an automated Python-based procedure followed by manual verification, yielding **236 unique articles**. These were screened independently by two reviewers based on titles and abstracts. Disagreements were resolved through discussion or consultation with a third reviewer.

Table 1: Search criteria

Search Criteria	Description
SC1: Radiotherapy-related terms	Radiotherapy OR Cancer treatment OR radiation therapy OR cancer
SC2: Organ segmentation-related terms	segmentation OR contour* OR organ segmentation OR organ delineation OR volume delineation
SC3: Machine learning-related terms	deep learning OR machine learning OR artificial intelligence OR convolutional neural networks OR CNN
SC4: Anatomical focus	pelvic area OR pelvis OR prostate OR cervical
SC5: Target volume-related terms	organ-at-risk OR CTV OR GTV OR OAR OR gross tumor volume OR clinical target volume

Table 2: Inclusion and exclusion criteria

Inclusion Criteria	Exclusion Criteria
IC1: Peer-reviewed journal articles ensuring methodological rigor.	EC1: Systematic reviews, conference proceedings, book chapters, and non-peer-reviewed sources.
IC2: Articles published in English to ensure consistency and accessibility.	EC2: Studies not focused on radiotherapy-related organ segmentation.

Articles meeting the inclusion criteria were assessed through full-text review. Studies were excluded if they did not focus on the pelvic region, did not address organ segmentation as a primary task, or corresponded to secondary research. Following this process, **173 articles** were included in the final analysis.

2.3. Analysis Categories

The selected studies were analyzed using a set of predefined categories to ensure a structured and consistent synthesis of the evidence. These categories were derived from the research questions and refined during the preliminary literature screening. The framework was designed to capture both the technical characteristics of the proposed segmentation approaches and methodological aspects influencing the reliability and clinical applicability of the reported results.

Two complementary groups of analytical dimensions were considered. The first group focused on the methodological and technical characteristics of the segmentation models and constitutes the basis of the descriptive analysis presented in the Results section. The second group comprised methodological indicators related to validation practices, reproducibility, and annotation quality, which were used primarily in the assessment of evidential reliability discussed in Section 4.6.

The main descriptive analysis categories are summarized in Table 3. These categories were used to systematically extract and organize information regarding model architectures, clinical targets, imaging inputs, evaluation strategies, optimization procedures, and dataset characteristics.

Table 3: Primary analysis categories used for data extraction and synthesis

Category	Description
Machine learning or deep learning methods and architectures	This category examines the machine learning (ML) and deep learning (DL) techniques employed for organ segmentation in radiotherapy. The analysis focuses on dominant architectures such as convolutional neural networks (CNNs), U-Net and its variants, V-Net, and transformer-based models, as well as extensions including attention mechanisms, hybrid architectures, and ensemble strategies.

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Category	Description
Cancer types addressed	This category explores the clinical contexts addressed by the reviewed studies, with emphasis on pelvic malignancies such as prostate, cervical, rectal, and bladder cancer, and considers how anatomical context may influence segmentation requirements and model evaluation.
Imaging modalities	This category analyzes the imaging modalities used as input for segmentation models, including CT, MRI, PET, and ultrasound, and examines their implications for soft-tissue representation, image contrast, and modality-specific segmentation challenges.
Evaluation metrics	This category reviews the quantitative metrics used to assess segmentation performance. These include overlap-based measures (e.g., Dice Similarity Coefficient and Jaccard Index), distance-based metrics (e.g., Hausdorff Distance and Average Surface Distance), and, when reported, clinically oriented indicators such as dosimetric parameters derived from dose–volume histograms.
Optimization algorithms and hyperparameter tuning	This category examines the optimization strategies used during model training, including gradient-based optimizers (e.g., Adam, SGD, RMSProp) and the reporting of hyperparameter tuning procedures or model selection strategies.
Dataset characteristics	This category evaluates dataset-related factors relevant to model development and generalization, including dataset size, institutional diversity, imaging heterogeneity, and annotation characteristics.

In addition to these descriptive categories, a set of methodological indicators was extracted to evaluate evidential reliability across studies. These indicators included validation setting (internal or external validation), dataset diversity (e.g., multi-centre datasets and scanner heterogeneity), reproducibility practices (e.g., availability of code or datasets), reporting of annotation procedures, and clinical or dosimetric evaluation. These variables were later used to characterize the evidential maturity of the field and to derive an overall evidential strength classification for each study.

3. Results

This section presents a structured synthesis of the studies that met the inclusion criteria. The analysis focuses on identifying major trends in deep learning approaches for organ segmentation in radiotherapy, including methodological choices, evaluation practices, and research evolution over time. Detailed per-study information is provided in the Supplementary Material.

3.1. Main Features of the Selected Articles

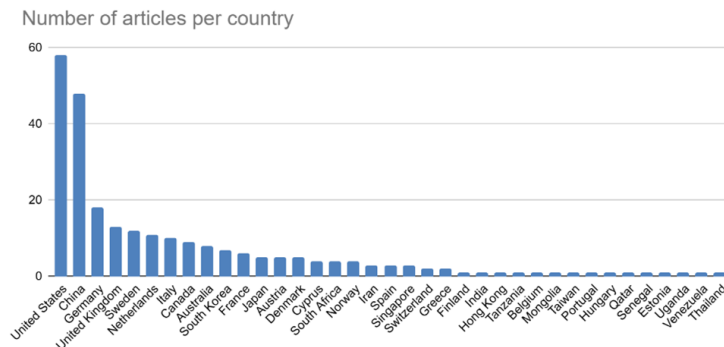
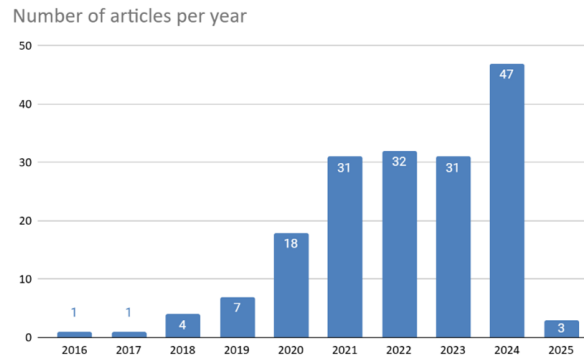


Figure 2: Number of articles per country.

Table 4: Distribution of segmentation methods by architectural category

Category	Studies (approx.)	Typical models	Trend
FCN (U-Net family)	Majority (>70%)	U-Net, 3D U-Net, nnU-Net	Dominant
CNN-based	Moderate (~20–30%)	DeepLabV3+, ResNet	Stable
GAN-based	Limited (<10%)	GAN, cGAN	Niche (augmentation)
Traditional ML	Marginal (<10%)	SVM, RF	Declining
Atlas-based	Limited (<10%)	Atlas-based	Hybrid use
Hybrid / specialized	Increasing	CE-Net, multi-task	Emerging
Commercial software	Moderate (~15–20%)	MIM, MVision	Clinical adoption

Figure 2 illustrates the geographic distribution of the selected studies. Research activity is concentrated in a small group of countries, led by the United States and China, with additional contributions from several European countries and Canada. This pattern indicates a broad but uneven international participation in the development of deep learning approaches for pelvic organ segmentation.

**Figure 3:** Number of articles per year.

The temporal distribution of publications (Figure 3) shows a rapid expansion of research activity after 2019, culminating in a peak of publications in 2024. This sustained growth reflects the consolidation of deep learning as a standard methodological framework for organ segmentation in radiotherapy. A small number of articles indexed as 2025 were also retrieved, likely due to early online publication.

3.2. Findings per Analysis Category

The findings were organized according to the main analytical categories of this review: segmentation methods and architectures, cancer types, imaging modalities, evaluation metrics, optimization strategies, and dataset characteristics. The complete per-study mapping is provided in the Supplementary Material (Table S1) and serves as the basis for the aggregated analyses presented below.

3.2.1. Machine learning or deep learning methods and architectures used

Among the 173 reviewed studies, 122 relied on a single segmentation method, 23 compared multiple approaches, and 30 employed commercial software. The identified methods were grouped according to their architectural characteristics to facilitate comparison. Table 4 summarizes the main method categories, their approximate prevalence, representative models, and general trends across the literature. A detailed per-method mapping is available in the Supplementary Material (Table S2).

The reviewed literature shows a clear predominance of fully convolutional architectures, particularly U-Net and its variants, which remain the dominant paradigm across segmentation tasks, imaging modalities, and datasets. CNN-based models continue to appear regularly but are increasingly incorporated into hybrid architectures that integrate attention mechanisms or transformer-based components to improve feature representation.

Table 5: Distribution of cancer types in the reviewed studies

Cancer type	Studies (n)
Pelvic area only (n = 153)	
Prostate	82
Cervical	54
Other pelvic cancers (bladder, anal, rectal, mixed cases)	17
Pelvic and other anatomical regions (n = 20)	
Prostate + other regions	14
Pelvic (unspecified) + other regions	5
Cervical + other regions	1
Total	173

Other approaches play more specialized roles. GAN-based methods are mainly used for data augmentation or domain adaptation rather than as primary segmentation models, while traditional machine learning techniques appear only marginally and mostly within hybrid or ensemble configurations. Atlas-based approaches continue to serve complementary purposes, often combined with deep learning methods to incorporate prior anatomical information.

Finally, the presence of commercial software solutions indicates a gradual translation of deep learning-based segmentation into clinical radiotherapy workflows.

This architectural distribution provides the basis for the broader analysis of methodological trends discussed in the following sections.

3.2.2. Types of cancer in the literature review

Table 5 summarizes the distribution of cancer types addressed in the reviewed studies. The literature is strongly concentrated on pelvic-only applications (153 of 173 studies), with prostate and cervical cancer accounting for the majority of cases. Prostate cancer was the most frequently investigated condition, followed by cervical cancer, while other pelvic malignancies were only marginally represented.

A smaller subset of studies (20 of 173) examined pelvic cancers together with other anatomical regions, most commonly in combination with head and neck, breast, or thoracic malignancies. These studies typically addressed multi-site segmentation problems or explored model performance across different anatomical domains, but remain relatively limited in number.

3.2.3. Types of images used in organ segmentation

Computed tomography (CT) was the predominant imaging modality in the reviewed studies, followed by magnetic resonance imaging (MRI) and cone beam CT (CBCT). Other modalities, including positron emission tomography (PET), transrectal ultrasound (TRUS), megavoltage CT (MVCT), and kilovoltage CT (kVCT), appeared only occasionally. The overall distribution of modalities is shown in Figure 4.

This distribution reflects the central role of CT in radiotherapy planning, while MRI remains the main complementary modality for improved soft-tissue delineation. CBCT is primarily used in image-guided and adaptive radiotherapy workflows.

To explore the relationship between imaging modality and segmentation strategy, a normalized comparison of modality-specific methods was conducted. The results are presented in Figure 5. The figure shows that U-Net and CNN-based architectures are the most consistently used approaches across modalities, particularly in CT and MRI studies. More specialized strategies, including multi-task learning, ensemble methods, and GAN-based approaches, appear relatively more frequently in MRI and PET studies, where image characteristics are often more complex or heterogeneous.

The results indicate that segmentation research remains strongly anchored in CT-based radiotherapy workflows, while gradually expanding toward modality-aware approaches, particularly in MRI, PET, and CBCT settings.

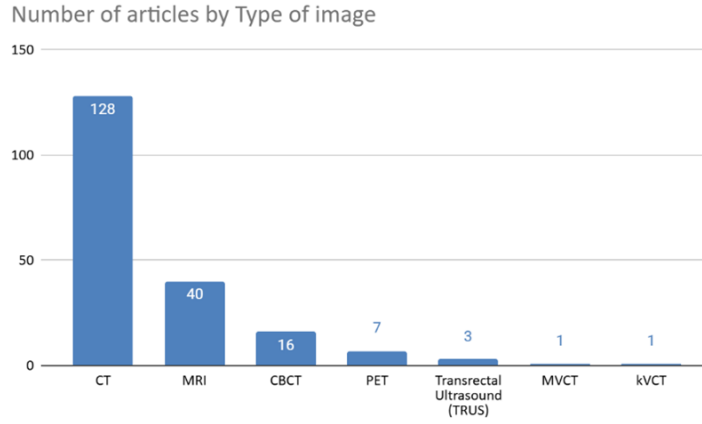


Figure 4: Frequency of imaging modalities used in the reviewed studies.

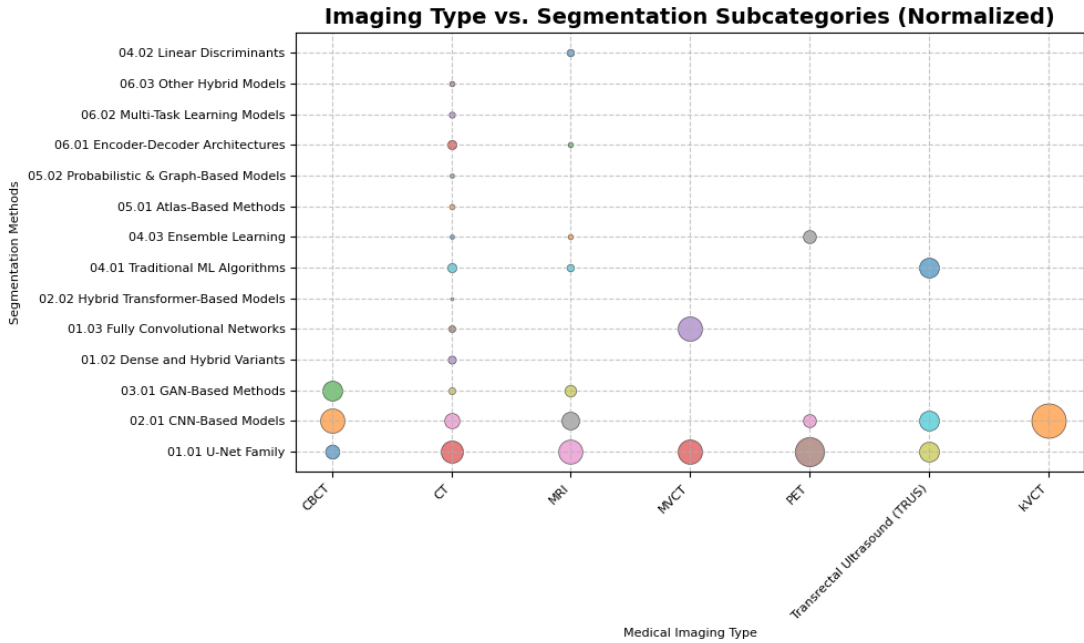


Figure 5: Normalized comparison of segmentation subcategories across medical imaging modalities. Bubble size represents the relative frequency of each method within each modality.

3.2.4. Evaluation metrics applied in the reviewed studies

The evaluation metrics reported in the reviewed studies were grouped into five broad categories: overlap-based, surface-based, classification, clinical and dosimetric, and efficiency-related measures. Figure 6 highlights the most frequently reported metrics, while Table 6 provides a complete overview of all metrics identified in the literature. Definitions of metric abbreviations are available in Supplementary Table S4.

Across the reviewed studies, geometric evaluation criteria were clearly predominant. Overlap-based and surface-based metrics accounted for the majority of reported measures, with the Dice Similarity Coefficient (DSC) being by far the most frequently used metric, followed by Hausdorff Distance (HD), the 95th percentile Hausdorff Distance (95HD), and Mean Surface Distance (MSD). This pattern indicates that segmentation quality is most commonly assessed in terms of spatial overlap and boundary agreement.

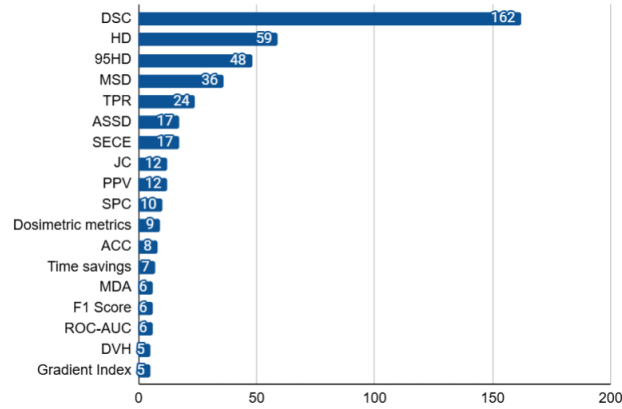


Figure 6: Frequency of evaluation metrics reported in the reviewed studies.

Classification metrics, including sensitivity, precision, specificity, accuracy, and F1 score, were reported less frequently and generally played a secondary role. Their presence reflects the interpretation of segmentation as a voxel-level classification task, although this perspective is not dominant in the literature.

Metrics directly related to clinical impact were comparatively rare. Dosimetric evaluation measures, DVH-based parameters, and plan quality indicators were reported in only a limited subset of studies. Similarly, workflow-oriented measures—such as clinician-based evaluation, time savings, and contour correction time—appeared much less often than geometric metrics. Together, these findings suggest that evaluation practices remain largely centered on geometric performance, while clinically oriented and workflow-related assessments are still relatively limited.

Table 6: Summary of evaluation metrics used in the literature. Counts indicate the number of articles in which each metric was reported; individual studies may use multiple metrics. Definitions of metric abbreviations are provided in Supplementary Table S4.

Metric	No. of articles
Overlap-based metrics	
Dice Similarity Coefficient (DSC)	162
Jaccard Coefficient (JC)	12
Relative Volume Difference (RVD)	1
Surface-based metrics	
Hausdorff Distance (HD)	59
95th Percentile Hausdorff Distance (95HD)	48
Mean Surface Distance (MSD)	36
Average Symmetric Surface Distance (ASSD)	17
Mean Distance to Agreement (MDA)	6
Root Mean Square Distance (RMSD)	3
Classification metrics	
Sensitivity / True Positive Rate (TPR)	24
Specificity (SPC)	10
Precision / Positive Predictive Value (PPV)	12
Accuracy (ACC)	8
F1 Score (F1)	6
Area Under the ROC Curve (ROC-AUC)	6
Clinical and dosimetric evaluation metrics	
Dosimetric evaluation metrics (DVH-derived parameters)	9
Dose–Volume Histogram parameters (DVH)	5

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Metric	No. of articles
Plan Quality Metrics (PQM)	2
Gradient Index (GI)	5
Efficiency and usability metrics	
Subjective Evaluation by Clinical Experts (SECE)	17
Time Savings (TS)	7
Contour Editing / Correction Time (CCT)	3
Turing Test–based evaluation (TT)	4
Correlation metrics (Pearson r , Spearman ρ)	3

3.2.5. Optimization methods identified in the literature review

Optimization practices were examined across three complementary dimensions: the training algorithms reported, the strategies used for hyperparameter optimization, and the hyperparameters explicitly documented in the reviewed studies. Together, these elements influence model convergence, generalization, and reproducibility.

Optimization algorithms

Table 7 summarizes the optimization algorithms reported in the reviewed studies. The distribution is highly concentrated, with Adam-based methods clearly dominating, followed by a limited use of SGD-based approaches, while alternative optimizers appear only sporadically. A substantial proportion of studies did not report the optimizer used.

This pattern indicates a strong reliance on adaptive first-order optimization with relatively little exploration of alternative strategies. The absence of optimizer reporting in many studies also raises transparency concerns, since optimization choices directly affect training dynamics and model performance.

Hyperparameter optimization methods

Explicit hyperparameter optimization strategies were reported only in a minority of studies. As summarized in Table 8, most articles did not describe any systematic optimization procedure, while manual tuning remained the most common reported approach. Structured search strategies, including grid search and Bayesian optimization, were comparatively rare.

These results suggest that hyperparameter optimization remains largely unsystematic across the literature, with relatively limited adoption of reproducible search strategies.

Reported hyperparameters

Reporting of individual hyperparameters was also uneven. Table 9 summarizes the most frequently documented groups of hyperparameters, showing that reporting is concentrated on a small set of core training variables.

Learning rate and batch size were the most frequently reported hyperparameters, followed by the number of epochs. Most other architectural, regularization, and optimization-related parameters appeared only sporadically. As a result, it is often difficult to determine whether reported performance reflects robust optimization procedures or a limited set of conventional training choices.

The results indicate that optimization reporting remains inconsistent and frequently incomplete. Adam is the predominant optimizer, systematic hyperparameter search is relatively uncommon, and reporting is largely limited to a small subset of standard training variables. This suggests that methodological transparency in optimization has not progressed at the same pace as architectural development, leaving optimization practices as a persistent source of variability in deep learning-based radiotherapy segmentation studies.

3.2.6. Dataset characteristics

Dataset size, measured as the number of patients, was strongly concentrated in the small- to medium-scale range (Figure 7). Most studies relied on fewer than 100 cases, with the 51–100 patient range being the most common, followed by cohorts of 1–50 patients. Studies using between 101 and 300 patients were also present, whereas large-scale datasets remained comparatively rare. Only a small subset of studies used more than 400 cases, and several did not report dataset size.

Table 7: Optimization algorithms reported in the reviewed studies. Counts indicate the number of studies in which each optimizer was mentioned.

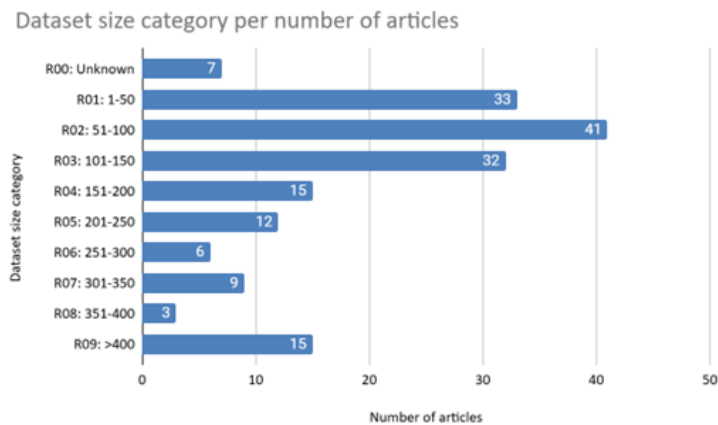
Optimization algorithm	Studies (n)
Adam and its variants	89
Not reported	68
SGD and its variants	18
Others	3

Table 8: Hyperparameter optimization methods reported in the reviewed studies.

Hyperparameter optimization method	Studies (n)
Not reported	108
Manual optimization	34
Systematic optimization (grid search, Bayesian, others)	31

Table 9: Most frequently reported hyperparameters in the reviewed studies

Category	Hyperparameter	Studies (n)
Training-related	Learning rate	90
	Batch size	64
	Number of epochs	25
	Other training parameters	10
Architectural	Number of layers	7
	Kernel size	4
	Other architectural parameters	4
Regularization	Dropout rate	11
	Loss-related parameters	14
	Other regularization strategies	3
Optimization-related	Patch size	6
	Other optimization parameters	11

**Figure 7:** Dataset size.

These results indicate that the literature remains largely based on relatively small training cohorts. Limited dataset size may increase the risk of overfitting and restrict the generalizability of reported models, while the scarcity of large and heterogeneous datasets continues to represent an important barrier to reproducibility and clinical translation.

4. Discussion

The results of this systematic review indicate that deep learning–based organ segmentation in pelvic radiotherapy has advanced rapidly in terms of architectural development, yet several methodological aspects remain comparatively underdeveloped. Across the 173 analyzed studies, U-Net–derived architectures dominate the literature, while limitations related to dataset scale, optimization practices, validation strategies, and reproducibility persist. These factors suggest that variability in reported segmentation performance often reflects differences in experimental design rather than purely algorithmic improvements.

The discussion interprets these findings by examining architectural trends, organ-specific segmentation performance, dataset characteristics, optimization strategies, evaluation practices, and evidential reliability. Together, these dimensions provide a clearer perspective on the current maturity of the field and the methodological challenges that must be addressed to support the reliable clinical deployment of automated segmentation systems in radiotherapy.

4.1. Architectural trends in deep learning for pelvic organ segmentation

A consistent finding across the reviewed literature is the sustained dominance of U-Net–based architectures in pelvic organ segmentation for radiotherapy. Approximately 76% of the analyzed studies relied on either 2D or 3D variants of U-Net ([27], [28], [29], [30], [31]). This predominance reflects the suitability of the encoder–decoder design with skip connections for radiotherapy segmentation tasks, where accurate delineation requires the integration of global anatomical context with precise boundary localization.

Architectural development in this domain has progressed largely through extensions of the U-Net paradigm rather than through entirely new model families. Residual and dense connectivity patterns have been incorporated into architectures such as ResUNet++ and Dense V-Net to facilitate deeper feature representations and improve gradient propagation ([27], [32]). In parallel, auto-configured frameworks such as nnU-Net demonstrated that segmentation performance depends not only on network topology but also on systematic optimization of preprocessing pipelines, training schedules, and hyperparameter configurations ([33]). These findings underscore the importance of methodological standardization alongside architectural design.

Another relevant line of development concerns architectures designed to enhance multi-scale contextual modeling. Methods such as DeepLabv3+ and RefineNet incorporate mechanisms that aggregate information across multiple receptive fields, a property that is particularly relevant in pelvic radiotherapy, where anatomical structures differ substantially in size, shape regularity, and boundary definition ([34], [35]). Such approaches aim to improve the representation of both local contour detail and broader anatomical context, especially in structures whose delineation depends on spatial relationships with adjacent tissues.

More recent work has explored attention mechanisms and transformer-based components to strengthen the modeling of spatial dependencies between pelvic structures. Several studies reported improved performance when channel-wise or spatial attention modules were integrated into convolutional architectures ([36], [35], [37]), and hybrid CNN–ViT models have shown promising results in multi-organ segmentation tasks ([38], [39]). Nevertheless, evidence supporting consistent advantages of transformer-based approaches across heterogeneous clinical datasets remains limited. The literature therefore suggests that, although architectural experimentation continues, most practical progress in pelvic radiotherapy segmentation still occurs within an expanded design space centered on U-Net variants, multi-scale feature aggregation, and increasingly hybrid architectures.

4.2. Organ-specific segmentation performance

Segmentation performance in pelvic radiotherapy varies substantially across anatomical structures, reflecting differences in organ morphology, imaging contrast, and contouring conventions. Although many studies report aggregated performance metrics across multiple organs, evaluating segmentation accuracy at the organ level provides a more informative view of the practical capabilities and limitations of current deep learning models.

Large organs-at-risk with well-defined anatomical boundaries consistently achieve the highest segmentation accuracy. The bladder represents the most stable example across the reviewed literature. Owing to its relatively homogeneous intensity and clearly defined boundaries in CT imaging, most studies report Dice Similarity Coefficients

Table 10: Typical segmentation performance across pelvic anatomical structures reported in the reviewed literature.

Organ / Structure	Typical DSC	Segmentation Difficulty	Key Anatomical Factors
Bladder	0.90–0.97	Low	Large volume, homogeneous intensity, clear CT boundaries
Femoral heads	0.93–0.98	Low	High contrast bone structure, stable morphology
Rectum	0.85–0.91	Medium	Variable filling, deformation, adjacency to soft tissue
CTV (prostate-related)	0.80–0.88	Medium	Clinically defined boundaries, contouring variability
Seminal vesicles	0.75–0.85	Medium–High	Irregular morphology, variable visibility in CT
Penile bulb	0.53–0.70	High	Small structure, weak contrast, limited voxel representation

above 0.90, with several analyses describing ranges between 0.92 and 0.97 depending on imaging modality and dataset characteristics ([29], [40]). Comparable behaviour is observed for the femoral heads, whose strong bone contrast and relatively stable morphology facilitate highly consistent delineation.

Intermediate performance levels are typically observed in structures such as the rectum and prostate-related target volumes. Rectal segmentation frequently reaches Dice values between 0.85 and 0.91 ([41]), although results vary due to anatomical deformation, variability in rectal filling, and the proximity of surrounding soft tissues. Clinical target volumes (CTV) introduce additional complexity because their boundaries are partially defined by treatment planning conventions rather than purely anatomical landmarks ([42]). Consequently, inter-observer variability becomes an important factor affecting both training data quality and model evaluation.

The greatest difficulties arise in small or poorly contrasted structures. Examples include the penile bulb and certain parametrial regions, which occupy a limited number of voxels and exhibit weak contrast relative to surrounding tissues. Several studies report Dice values between 0.53 and 0.70 for the penile bulb ([43], [44]), illustrating the challenge of learning stable representations for small anatomical targets within CT-based radiotherapy datasets. Similar limitations have been described for other small pelvic structures where partial volume effects and anatomical variability complicate both manual contouring and automated segmentation.

Table 10 summarizes typical segmentation performance ranges observed across key pelvic structures reported in the reviewed literature. The table illustrates how segmentation difficulty generally increases as anatomical structures become smaller, less contrasted, or more dependent on clinician-defined boundaries.

These organ-specific differences have important implications for interpreting segmentation metrics. Average performance values reported across multiple organs may conceal substantial variability between structures with markedly different segmentation difficulty. In radiotherapy planning, where contour accuracy directly influences dose distribution, these discrepancies are clinically significant. Future research should therefore place greater emphasis on improving robustness for anatomically challenging structures rather than focusing exclusively on incremental improvements in global Dice scores.

4.3. Dataset characteristics and generalizability

Dataset characteristics play a central role in determining the performance and robustness of deep learning models for pelvic organ segmentation. Imaging modality, dataset size and diversity, and annotation quality collectively define the conditions under which segmentation algorithms are trained and evaluated, and therefore strongly influence their capacity to generalize across clinical environments.

CT remains the predominant imaging modality in the reviewed literature, reflecting its central role in radiotherapy planning ([45], [46]). Nevertheless, MRI-based models often achieve superior performance for soft-tissue structures because of the higher contrast resolution of MRI, particularly for organs such as the prostate and bladder ([35], [40]). These observations highlight the importance of modality-specific imaging characteristics in segmentation performance. Hybrid strategies combining CT and MRI information have also been explored ([47], [48]), although their adoption remains limited due to the practical challenges associated with assembling multimodal datasets and performing accurate image registration.

Dataset size and institutional diversity represent additional determinants of model generalizability. The studies included in this review exhibit considerable variation in dataset scale, ranging from very small training cohorts to large multi-institutional collections ([49], [50], [51]). Larger and more heterogeneous datasets tend to produce more stable segmentation performance and improved cross-institutional generalization ([46], [52]). In contrast, models trained on small single-center datasets frequently show reduced performance when evaluated on external cohorts ([53], [54]). These findings are consistent with broader observations in medical imaging that limited training diversity increases the risk of overfitting and reduces model robustness in real clinical deployments.

Annotation quality constitutes another major source of variability. Manual delineation by radiation oncologists remains the dominant annotation strategy in most studies, ensuring clinical relevance but also introducing inter-observer variability. This issue is particularly evident for structures whose boundaries depend partly on clinical interpretation rather than purely anatomical landmarks. Clinical target volumes represent a typical example, as contour definitions vary across institutions and treatment planning protocols ([42], [55], [39]). Multi-center analyses have shown that inconsistent contouring guidelines can produce reductions in DSC of up to 3% ([52]), indicating that annotation variability can place an upper bound on the apparent performance of automated segmentation systems. Efforts to standardize contouring protocols through consensus guidelines have shown measurable improvements in segmentation performance ([31]).

Several methodological strategies have been proposed to mitigate limitations associated with dataset size and variability. Data augmentation remains one of the most widely adopted techniques for increasing the effective diversity of training samples ([49], [29]). Transfer learning has also been used to stabilize training when only limited datasets are available ([27], [35], [53]), and in some cases has enabled high segmentation performance even with fewer than twenty training cases in MRI-based studies ([56]). More recently, synthetic data generation and federated learning have been explored as mechanisms for expanding training resources while addressing institutional data-sharing constraints ([57], [58], [44]). These approaches reflect a growing recognition that improving dataset diversity may be as important as architectural innovation for advancing segmentation performance.

The evidence reviewed here indicates that dataset characteristics exert a substantial influence on segmentation performance, often comparable to that of architectural design choices. Continued progress in pelvic radiotherapy segmentation will therefore depend on expanding dataset diversity, harmonizing annotation practices, and establishing collaborative multi-institutional data resources capable of capturing the variability present in real-world clinical imaging.

4.4. Optimization practices

The performance of deep learning models depends not only on architectural design but also on the optimization strategies used during training. The reviewed literature indicates that optimization practices remain relatively homogeneous and are often insufficiently documented.

Across the 173 analyzed studies, the Adam optimizer emerged as the dominant training algorithm, appearing in approximately 49% of the reviewed works (e.g., [59], [49], [27], [38], [29]). Its adaptive learning rate and robustness to gradient variability likely explain its widespread adoption in segmentation tasks characterized by class imbalance and heterogeneous anatomical structures. In contrast, stochastic gradient descent (SGD), typically combined with momentum, was reported in only about 10% of the studies (e.g., [50], [30]). Comparative analyses suggest that Adam may provide modest improvements in segmentation performance relative to SGD under comparable training conditions ([60], [61]). Other optimizers, including RMSProp ([60]) and AdamW ([62]), were only sporadically reported, and systematic comparisons across optimization algorithms remain largely absent from the literature.

Hyperparameter optimization practices exhibit even greater methodological variability. Manual tuning was the most frequently reported approach, appearing in approximately 41% of the studies (e.g., [59], [63], [38]), whereas structured strategies such as grid search or other systematic optimization methods were reported in only about 7% of the literature (e.g., [27], [60]). Notably, roughly half of the reviewed studies did not explicitly describe the procedures used to determine training parameters (e.g., [64], [65], [66]), limiting reproducibility and hindering reliable comparisons between methods.

Among the hyperparameters most frequently reported, learning rate and batch size received the greatest attention. Learning rate adjustments were described in approximately ninety studies (e.g., [59], [27], [60]), reflecting their central role in training stability and convergence behaviour. Batch size, reported in sixty-four studies, was often constrained by GPU memory limitations, particularly in three-dimensional architectures ([67], [68]). Other parameters, including

Table 11: Evaluation metrics and clinical validation approaches reported in the reviewed literature.

Category	Metric / Evaluation	Approximate Frequency	Primary Interpretation or Outcome
<i>Geometric segmentation metrics</i>			
	Dice Similarity Coefficient (DSC)	>93%	Volumetric overlap between predicted and reference contours
	Hausdorff Distance (HD / HD95)	~30–35%	Maximum or near-maximum boundary deviation between contours
	Surface distance metrics (MSD, ASSD)	<45%	Average surface alignment between predicted and reference structures
<i>Clinical evaluation approaches</i>			
	Dose-volume histogram (DVH) analysis	~12%	Assessment of treatment planning constraints and dosimetric impact
	Contour editing time	~10%	Reduction in manual correction workload and workflow efficiency
	Expert qualitative scoring	~22%	Clinical acceptability assessed through blinded expert review

dropout rates, loss weighting strategies, and patch size, were less consistently explored but occasionally produced substantial performance gains in specific tasks ([69]).

Only a small subset of studies investigated more advanced optimization strategies, such as reinforcement learning–based hyperparameter tuning ([70]) or automated configuration frameworks. As a result, optimization procedures are frequently treated as secondary implementation details rather than as methodological components that can substantially influence segmentation outcomes.

Overall, the evidence suggests that optimization practices in pelvic organ segmentation research remain largely pragmatic and insufficiently standardized. Greater transparency in reporting training configurations, together with more systematic evaluation of hyperparameter optimization strategies, would substantially improve the reproducibility and comparability of future studies.

4.5. Evaluation practices

Reliable evaluation of segmentation performance is essential for determining the clinical readiness of deep learning models in radiotherapy applications. The studies analyzed in this review show a strong methodological convergence toward a limited set of geometric performance metrics, particularly the Dice Similarity Coefficient (DSC) and the Hausdorff Distance (HD). While these metrics provide valuable information about volumetric overlap and boundary deviation, their relationship to clinically meaningful outcomes remains an active topic of discussion.

Table 11 summarizes the evaluation approaches reported in the reviewed literature, distinguishing between geometric segmentation metrics and clinically oriented validation strategies. Evaluation practices remain strongly dominated by geometric indicators of segmentation accuracy, whereas direct assessments of clinical impact appear less frequently.

Among geometric metrics, the Dice Similarity Coefficient is by far the most widely reported indicator of segmentation performance, appearing in more than 93% of the reviewed studies (e.g., [49], [38], [29], [67]). Its popularity derives from its intuitive interpretation and its ability to provide a standardized measure of volumetric agreement between predicted and reference contours. In pelvic radiotherapy applications, DSC thresholds of approximately 0.85 for organs-at-risk and around 0.70 for clinical target volumes are often considered indicative of acceptable segmentation accuracy (e.g., [64], [43]). However, overlap-based metrics may conceal clinically relevant boundary discrepancies. For example, prostate CTV segmentations achieving DSC values near 0.89 have been reported to require manual revision due to localized contour deviations near adjacent structures ([42]).

Boundary-based measures such as the Hausdorff Distance or its 95th percentile variant (HD95) are frequently used to complement overlap metrics by quantifying local geometric discrepancies between contours. These metrics provide a more sensitive assessment of boundary accuracy, particularly in regions where small spatial deviations may influence treatment planning. For instance, HD95 values below 5 mm for bladder segmentation have been associated

with minimal variation in key dose-volume histogram parameters ([67]). Nevertheless, discrepancies between overlap and boundary metrics remain common, as illustrated by bladder segmentations achieving DSC values above 0.90 while still presenting localized boundary errors exceeding several millimeters ([71]).

Surface-based metrics such as Mean Surface Distance (MSD) and Average Symmetric Surface Distance (ASSD) have been proposed to further characterize anatomical boundary alignment. These measures quantify average surface discrepancies and therefore provide complementary information regarding segmentation accuracy. Despite their potential advantages, they remain less consistently reported across studies (e.g., [40], [72]), indicating that evaluation practices in pelvic organ segmentation continue to rely primarily on traditional geometric indicators.

In contrast to the widespread use of geometric metrics, relatively few studies directly evaluate the clinical consequences of segmentation errors. As summarized in Table 11, only a minority of works assess downstream radiotherapy outcomes such as dose-volume histogram compliance, manual contour editing time, or qualitative clinical acceptance. DVH-based analyses were reported in approximately 12% of the studies (e.g., [67], [43], [73]) and provide a direct link between segmentation accuracy and treatment planning quality. In one investigation, segmentation models achieving DSC values above 0.80 and HD95 values below 6 mm for high-risk clinical target volumes produced treatment plans comparable to those generated using expert contours ([46]).

Operational metrics related to clinical workflow are also beginning to appear in the literature. Several studies reported reductions of up to 47% in manual contouring time when automated segmentation tools were integrated into radiotherapy planning pipelines ([74]). Qualitative evaluation by clinical experts has likewise been adopted as a complementary validation strategy, with approximately 22% of the reviewed studies incorporating blinded expert scoring of automated contours (e.g., [75], [76]). In some cases, automated segmentations were considered comparable to, or preferable to, manual delineations.

The evaluation practices identified in this review suggest a gradual shift toward more comprehensive validation frameworks. Nevertheless, the continued reliance on overlap-based metrics indicates that evaluation methodologies have not yet fully converged toward clinically grounded standards. Future work will benefit from evaluation protocols that integrate geometric accuracy, boundary agreement, dosimetric impact, and workflow efficiency in order to more accurately reflect the clinical value of automated segmentation systems in radiotherapy planning.

4.6. Study quality and reproducibility considerations

Assessing the reliability of the available evidence is essential for interpreting the reported performance of deep learning models in medical image analysis. In this context, high segmentation accuracy does not necessarily imply methodological robustness or clinical readiness. The credibility and translational potential of reported results depend on several factors, including validation strategies, dataset characteristics, implementation transparency, and the reliability of the ground-truth annotations used during model development and evaluation..

To examine these aspects, a structured assessment framework was applied using indicators grouped into three methodological domains: validation and data-source characteristics, reproducibility and transparency, and clinical relevance with optimization reporting. These indicators were extracted systematically during the review process and are summarized in Supplementary Material Table S3. Figure 8 presents the proportion of studies satisfying each evidential criterion.

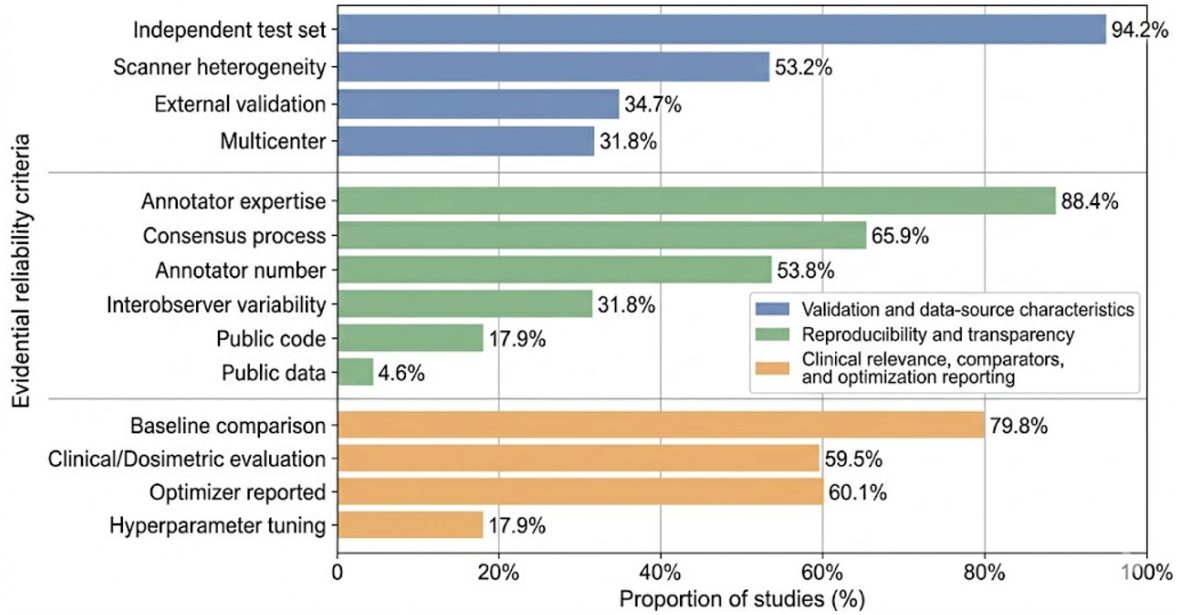


Figure 8: Coverage of evidential reliability criteria across the reviewed studies. The bars represent the proportion of studies satisfying each methodological criterion related to model validation, dataset characterization, annotation quality, and reproducibility.

4.6.1. Generalization and external validity

The ability of deep learning models to generalize beyond the data used during development is a fundamental requirement for clinical translation. Models trained and evaluated within a single institutional dataset may achieve high segmentation performance while failing to maintain similar accuracy across different hospitals, scanners, or acquisition protocols.

Most studies incorporate some form of independent test-set evaluation. Specifically, 94.2% of the reviewed articles report the use of a test set separated from the training data. However, validation strategies remain predominantly internal. A majority of studies (63.0%) rely exclusively on internal validation, while only 3.5% report purely external validation and 31.2% combine internal and external datasets. Consequently, only about one third of the literature evaluates model performance using data from independent institutions.

Dataset diversity shows a similar pattern. Although 53.2% of studies include some degree of scanner or acquisition heterogeneity, only 31.8% rely on multi-center datasets. As a result, many models are evaluated under conditions that may not fully represent the variability encountered in real clinical environments. These findings suggest that, despite relatively mature experimental workflows, robust external validation remains limited across the literature.

4.6.2. Reproducibility and transparency

Reproducibility is a central requirement for the verification and cumulative development of machine learning research. In medical image analysis, access to implementation code, trained models, and datasets enables independent replication of results and facilitates objective comparison between competing methods.

Across the reviewed literature, reproducibility practices remain limited. Only 17.9% of the studies provide access to their implementation code (31 out of 173 articles), whereas 53.8% explicitly report that the code is not available and 28.3% provide no information regarding code accessibility.

Dataset accessibility is even more restricted. Only 4.6% of the studies rely on publicly available datasets that allow independent reproduction of the reported experiments. A further 28.3% indicate that data may be obtained upon request, while 58.4% explicitly report that the datasets are not publicly accessible. This limited availability likely reflects the privacy and regulatory constraints associated with clinical imaging data, which often restrict public data sharing despite its importance for benchmarking and reproducibility.

These observations highlight a substantial gap between conventional experimental evaluation practices and practical reproducibility. Although most studies adopt standard training–validation–testing workflows, limited access to

code and datasets restricts independent verification of reported results and complicates systematic comparison between segmentation methods.

4.6.3. Reliability and reporting of ground-truth annotations

The reliability of ground-truth annotations directly influences both model training and the interpretation of segmentation performance. In supervised learning frameworks, the quality of the reference contours effectively determines the upper bound of achievable model accuracy.

Most studies report at least basic information regarding annotator expertise. In particular, 88.4% of the reviewed articles indicate that annotations were produced by expert clinicians, typically radiation oncologists or experienced radiologists.

Other aspects of annotation reliability are less consistently documented. Only 53.8% of studies report the number of annotators involved in the contouring process, and 65.9% describe the use of consensus procedures to resolve disagreements between observers. Explicit quantification of interobserver variability is reported in only 31.8% of the reviewed articles.

This limited reporting is particularly relevant in pelvic organ segmentation, where anatomical boundaries may be difficult to define consistently and contouring practices vary across clinicians and institutions. When interobserver differences are not quantified, the uncertainty associated with reference annotations may remain unaccounted for during model evaluation.

4.6.4. Clinical relevance and methodological rigor

Methodological rigor also depends on the extent to which experimental evaluations reflect clinically meaningful conditions. In radiotherapy planning, this includes comparison with appropriate baselines, assessment of clinical or dosimetric impact, and transparent reporting of model optimization procedures.

Most studies compare their proposed methods against existing segmentation approaches. Specifically, 79.8% of the reviewed articles include baseline comparisons, typically involving widely used reference architectures such as U-Net variants or related convolutional neural network models ([29], [59]).

Clinical relevance is less consistently addressed. Only 59.5% of the reviewed studies report any form of clinical or dosimetric evaluation beyond geometric segmentation metrics. Some investigations assess the potential impact of automated contours on treatment planning parameters or include qualitative expert evaluations of contour acceptability ([77], [78]), whereas the remaining studies rely exclusively on geometric indicators such as Dice similarity coefficient or Hausdorff distance.

Reporting of model optimization practices shows a similarly heterogeneous pattern. The optimization algorithm used for network training is reported in 60.1% of the studies, while explicit descriptions of hyperparameter tuning strategies appear in only 17.9% of the literature.

4.6.5. Evidential maturity of the field

The indicators analyzed in this review suggest that deep learning research for pelvic organ segmentation in radiotherapy has reached a moderate level of evidential maturity. Independent test sets and comparative benchmarking against existing architectures are widely implemented and support the internal validity of many reported segmentation results.

When the methodological indicators are aggregated into an overall evidential strength score, 54.9% of the reviewed articles fall into the moderate evidential strength category, typically reflecting studies that adopt standard experimental workflows but lack elements such as external validation or reproducibility resources. In addition, 31.2% of studies were categorized as lower evidential strength, while only 13.9% satisfied the criteria for higher evidential strength.

Several elements required for robust clinical translation remain inconsistently implemented. External multi-institutional validation, reproducibility practices such as code and dataset availability, and systematic characterization of annotation variability are reported only in a minority of studies. These observations indicate that, although methodological development in the field is active, further improvements in validation design, reproducibility standards, and clinically grounded evaluation will be necessary to strengthen the evidential foundation of automated segmentation systems for radiotherapy planning.

4.7. Challenges and future research directions

Synthesizing the limitations identified across the reviewed corpus reveals that, while deep learning-based segmentation has matured beyond the proof-of-concept stage, several systemic barriers continue to impede its widespread

clinical adoption. These obstacles arise from a convergence of restricted data regimes, inherent anatomical ambiguity, variability in reference standards, and a persistent lack of validation under realistic clinical conditions.

Dataset scarcity and the prohibitive burden of expert annotation remain the most pervasive constraints, cited by 35% of the reviewed studies (62/173). The development of robust models is frequently hindered by small cohort sizes, a lack of institutional diversity, and the resource-intensive nature of manual contouring. As noted by [28], the logistical difficulties of data export and the limited scale of available datasets restrict the depth of both training and validation. These data-related bottlenecks are further exacerbated by class imbalance ([79]) and a reliance on single-center datasets ([80], [81]), which heighten the risk of overfitting and limit the model's ability to handle the domain shifts encountered in external cohorts. Consequently, cross-institutional generalization remains a critical concern for 28% of the literature, where variations in imaging protocols and local contouring conventions can lead to significant performance degradation ([82], [42]). These patterns suggest that a substantial portion of the field still operates under narrow experimental conditions that do not fully reflect real-world clinical variability.

Beyond data availability, the intrinsic anatomical complexity of the pelvic region presents significant hurdles for precise boundary delineation. Approximately 24% of the studies identify low-contrast interfaces, significant organ deformation, and tissue heterogeneity as primary sources of segmentation error. In CT-based radiotherapy, the lack of distinct signal intensity between target volumes and adjacent soft tissues often leads to semantic ambiguity. For instance, [83] highlighted the difficulty of resolving the clinical target volume (CTV) against surrounding structures, while [36] and [54] emphasized the challenges posed by the intraprostatic urethra and the rectum, respectively, due to their proximity to tissues with similar intensity profiles. These anatomical difficulties are intrinsically linked to inter-observer variability, an issue discussed by 37% of the authors. As demonstrated by [84] and [85], the absence of a “perfect” ground truth not only complicates model training but also introduces uncertainty in the evaluation of model performance. In supervised learning frameworks, this inherent label noise effectively sets a ceiling on the achievable geometric accuracy of automated systems.

The translational gap between high geometric performance and practical clinical utility constitutes a final significant barrier, addressed by 21% of the reviewed studies. There is a growing recognition that high Dice Similarity Coefficient (DSC) values do not always translate into clinical acceptability. Several investigations emphasize that automated contours still necessitate time-consuming manual refinements before they can be safely used in treatment planning ([67], [65]), while others point to a lack of interoperability with existing planning systems as a major hurdle ([56]). Although computational constraints such as hardware requirements or memory limitations are mentioned in 15% of the studies ([86], [87]), these are increasingly secondary to the need for validation in adaptive or time-sensitive radiotherapy workflows, where the speed and reliability of the segmentation must meet strict clinical windows.

These challenges have shaped the research priorities proposed for the coming years. Nearly half of the literature (89/173) advocates for the creation of larger and more diverse datasets through multi-center collaborations and the implementation of standardized annotation protocols ([88], [89]). Another prominent direction, supported by 73 studies, is the enhancement of algorithmic robustness in anatomically challenging scenarios ([49], [90], [91]). Furthermore, 30% of the corpus (56 studies) suggests a move toward multimodal or multi-organ frameworks, leveraging CT–MRI or PET data to resolve anatomical ambiguities ([45], [92], [93], [94]). Finally, 45 studies identify integration into clinical workflows, particularly for real-time and adaptive radiotherapy, as the most critical step for the next generation of segmentation models ([49], [82], [95]).

The evidence suggests that the field is transitioning from a period of rapid architectural experimentation toward a more mature phase focused on methodological transparency, data diversity, and validation frameworks that prioritize clinical impact over purely geometric metrics.

5. Conclusions

This systematic review of 173 peer-reviewed studies characterizes the current state of deep learning for pelvic organ segmentation, a domain defined by high anatomical complexity and stringent requirements for radiotherapy planning. The evidence synthesized here indicates that, although the field has achieved considerable technical maturity, important barriers remain for translating these models into robust and autonomous clinical tools.

The prevailing architectural paradigm remains centered on U-Net and its derivatives, which have proven effective in capturing both global anatomical context and fine boundary details in pelvic structures. The widespread adoption of residual connections, attention mechanisms, and hybrid CNN–Transformer configurations reflects a sustained period of architectural refinement that has pushed geometric accuracy toward near-expert levels for several organs-at-risk.

In parallel, the diversification of imaging inputs beyond CT, including MRI and CBCT, highlights growing efforts to improve soft-tissue delineation and to support adaptive radiotherapy workflows.

Despite these advances, this review identifies a persistent generalizability gap arising from structural limitations in current research practices. Although independent test-set evaluation is now common, most studies continue to rely on single-center datasets and internal validation strategies. The limited use of external validation across independent institutions and the scarcity of multi-center datasets indicates that the robustness of many reported models under real-world clinical variability has yet to be fully established. In addition, limited transparency in reproducibility—reflected by the low availability of code and datasets—restricts cumulative progress and complicates objective benchmarking of competing methods.

Another key observation is the disconnect between geometric performance and clinical utility. The widespread reliance on overlap-based metrics such as the Dice Similarity Coefficient often obscures the potential clinical impact of segmentation errors on dosimetric distributions and clinician workload. Furthermore, the inconsistent reporting of annotation protocols and inter-observer variability suggests that the uncertainty associated with reference contours is frequently underestimated during model evaluation.

Future progress in pelvic radiotherapy segmentation will therefore depend less on incremental architectural modifications and more on strengthening the evidential foundations of the field. The development of large multi-institutional datasets, adoption of reproducible research practices, and evaluation frameworks incorporating clinically meaningful outcomes will be essential steps toward translating automated segmentation from an experimental tool into a reliable component of routine radiotherapy planning and adaptive treatment workflows.

Author Contributions

Conceptualization, I.D.; methodology, I.D.; formal analysis, I.D., G.C. and R.S.; investigation, I.D., G.C. and R.S.; data curation, I.D.; writing—original draft preparation, I.D., G.C. and R.S.; writing—review and editing, I.D., G.C. and R.S.; visualization, I.D., G.C. and R.S.; supervision, I.D., G.C. and R.S.; project administration, I.D., G.C. and R.S. All authors have read and agreed to the published version of the manuscript.

Declaration of Competing Interest

The authors declare no conflict of interest.

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Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this manuscript, the authors used ChatGPT (OpenAI) to assist with language editing and improvement of readability. The tool was used solely to refine the presentation of the text. All scientific content, interpretations, and conclusions were developed by the authors. The authors reviewed and edited the generated output as necessary and take full responsibility for the content of this publication.

Supplementary Material

The supplementary material provides additional information supporting the analyses presented in this systematic review. It includes detailed tables that complement the aggregated results reported in Section 3 and Section 4.

S1. Per-study analytical mapping

Table S1 presents the detailed mapping of all selected studies according to the analytical categories defined in the data extraction process. This table contains the full per-study classification used to derive the aggregated analyses discussed in Section 3.

Table S1: Findings by analysis category in selected articles

ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
1	Gustafsson et al. [59]	2021	InceptionResNetV2	CT; MRI	Prostate	ACC; F1 Score	299
2	Bordigoni et al. [45]	2024	Commercial Software	CT	Cervical; Prostate	DSC; HD; DAP	80
3	Breto et al. [49]	2022	Mask R-CNN	MRI	Cervical	DSC; HD; r	15
4	Liu et al. [63]	2020	DpnU-Net	CT	Cervical	DSC; 95HD; SECE	237
5	Min et al. [27]	2023	ResUNet++	MRI	Prostate	TPR; ROC-AUC; FPR	55
6	Arjmandi et al. [38]	2024	CNN-ViT	CT	Prostate	DSC; HD; MSD; RMSD	104
7	Eppenhof et al. [50]	2020	CNN	MRI	Prostate	DSC; HD; CD	5
8	Shen et al. [96]	2023	CNN	CT	Prostate	DSC; 95HD; SECE	217
9	Min et al. [82]	2021	3D U-Net	MRI	Prostate	DSC; ROC-AUC; SPC; TPR	Not specified
10	Kawula et al. [28]	2023	3D U-Net	MRI	Prostate	DSC; HD; MSD	92
11	Holzschuh et al. [47]	2023	3D U-Net	PET; CT	Prostate	DSC; TPR; SPC	180
12	Pirrone et al. [97]	2022	Ensemble machine learning (EML); Manual contouring	PET; CT	Prostate	TPR; SPC; ACC; ROC-AUC	43
13	Ma et al. [90]	2022	VB-Net	CT	Cervical	DSC; HD; MSD	535
14	Wang et al. [98]	2022	CNN	CT	Cervical	DSC; MSD; JC; ρ ; DM	375
15	Hobbis et al. [42]	2023	3D U-Net	CT	Prostate	DSC; HD; MSD	109
16	Kostyszyn et al. [60]	2021	3D U-Net	PET	Prostate	DSC; HD; ASSD; TPR; SPC	209
17	Wong et al. [64]	2021	Commercial Software	CT	Central nervous system (CNS); Head and neck; Prostate	DSC; 95HD; SECE	551

Continued on next page

ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
18	Kim et al. [69]	2024	DCT-UNet	MRI	Cervical	DSC; IoV	244
19	Cubero et al. [36]	2023	3D U-Net; nnU-Net	CT; MRI	Prostate	CLD	351
20	Xiao et al. [34]	2022	RefineNet; RefineNetPlus3D; FCN; U-Net; Context Encoder Network (CE-Net); U-Net3D; ResU-Net3	CT	Cervical	DSC; ASSD; JC	313
21	Kiljunen et al. [99]	2020	Commercial Software	CT	Prostate	DSC; IoV	30
22	Liu et al. [75]	2021	DpnU-Net	CT	Cervical	DSC; 95HD; SECE; TT	257
23	Balagopal et al. [29]	2021	3D U-Net	CT	Prostate	DSC; SECE	340
24	Rhee et al. [46]	2020	CNN	CT	Cervical	DSC; HD; MSD	2254
25	Wong et al. [100]	2020	Commercial Software	CT	Central nervous system (CNS); Head and neck; Prostate	DSC; HD	60
26	Wang et al. [101]	2024	CNN	CBCT	Cervical	DSC; 95HD; FPR; DM	21
27	Elguindi et al. [35]	2019	DeepLabV3+	MRI	Prostate	DSC	100
28	Wen et al. [102]	2024	U-Net	CT	Rectal; Prostate; Cervical	DSC; 95HD; ASSD; SECE	160
29	Nguyen et al. [103]	2019	U-Net	CT	Prostate	DSC; MSE	88
30	Kawula et al. [67]	2022	3D U-Net	CT	Prostate	DSC; HD; 95HD; DVH; GI	69
31	Conroy et al. [104]	2021	Atlas-based	CT	Prostate	DM; GI	25
32	Zhang et al. [105]	2021	CNN; Markov random field (MRF)	CT	Prostate	DSC; HD; VD	140
33	Kawula et al. [106]	2023	3D U-Net	MRI	Prostate	DSC; HD; 95HD	92
34	Mao et al. [77]	2020	CNN	CT	Prostate	TS; DVH	71
35	Berenato et al. [70]	2024	Commercial Software	CT	Prostate	DSC; HD; DM	20
36	Aström et al. [107]	2024	Not specified; Manual contouring	CBCT	Bladder	DSC; DM	10
37	Stenhouse et al. [81]	2024	RF; GBM	MRI	Cervical	IoV	127
38	Yeap et al. [57]	2023	ResNet-50	CT; CBCT	Prostate	DSC; RMSE	20
39	Jeong et al. [30]	2024	U-Net	CT	Prostate	DSC; HD; MSD	18
40	Tegtmeier et al. [78]	2024	CNN	CBCT	Prostate	DSC; 95HD; MSD	141
41	Cha et al. [56]	2021	DeepLabV3+	MRI	Prostate	DSC; APL	173

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ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
42	Zimmermann et al. [108]	2021	cGAN	MRI	Prostate	DSC; HD; GI	12
43	Doolan et al. [109]	2023	Commercial Software	CT	Breast; Head and neck; lung; Prostate	DSC; HD; APL; TS	80
44	Kibudde et al. [110]	2024	Commercial Software	CT	Head and neck; Prostate	DSC; MSD; 95HD	10
45	Sarria et al. [111]	2024	Commercial Software	CT	Head and neck; Breast; Prostate	DSC; TS	75
46	Koutoulakis et al. [79]	2023	U-Net	MRI	Prostate	DSC; HD	103
47	Lee et al. [112]	2023	FC-DenseNet; U-Net	MVCT	Prostate	DSC; MSD	100
48	Yang et al. [41]	2024	DeepLabV3+	MRI; CT	Prostate	DSC; ρ	40
49	Byrne et al. [113]	2022	Commercial Software	CBCT	Prostate	PQM	18
50	Balagopal et al. [48]	2024	U-Net	CT; MRI	Prostate	DSC; MSD; 95HD	56
51	Chung et al. [85]	2023	U-Net	CT	Cervical	DSC; 95HD; IoV; TT; CCT	180
52	Zabel et al. [114]	2021	U-Net; Atlas-based	CT	Prostate	DSC; TS; DM	15
53	Fransson et al. [115]	2024	U-Net	MRI	Prostate	DSC; DM	35
54	Balagopal et al. [116]	2021	PSA-Net	CT	Prostate	DSC; HD; MSD; ASSD; TPR; SPC	373
55	Chang et al. [117]	2021	3D U-Net	CT	Cervical	DSC	136
56	Fransson et al. [118]	2022	U-Net	MRI	Prostate	DSC; APL	17
57	Arabi et al. [119]	2018	DCNN	MRI	Prostate	DSC; MAE; GI	39
58	Urago et al. [120]	2021	Atlas-based; U-Net	CT	Prostate; Head and neck	DSC; HD; MDA	51
59	Chlap et al. [121]	2024	U-Net	CT	Gastric; Prostate	DSC; HD	60
60	Liu et al. [122]	2022	U-Net; cGAN	CBCT; CT	Prostate	ACC; MAE	91
61	Chen et al. [123]	2023	DCNN	CT; kVCT	Head and neck; Pelvic area	DSC	123
62	Jenkins et al. [55]	2021	Commercial Software	CT	Prostate	Other	232
63	Macomber et al. [124]	2018	Commercial Software	CT	Prostate	DSC; HD	193
64	Wang et al. [125], [51]	2023	Multi-task Learning	CT	Prostate	DSC; PPV; TPR; Other	Not specified

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ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
65	Rhee et al. [51]	2022	SVM	CT	Cervical	DSC; HD; MSD; 95HD; Other	87
66	Duan et al. [92]	2023	Commercial Software	CT	Prostate	DSC; HD; MSD	215
67	Holzschuh et al. [126]	2024	nnU-Net	PET	Prostate	DSC; HD; VC	161
68	Azzarouali et al. [127]	2024	Commercial Software	CBCT	Bladder	DSC; MDA; Other	17
69	Kaushik et al. [128]	2024	Not specified	CBCT; CT	Prostate	DSC; CES; Other; GI	71
70	Chan et al. [58]	2024	cGAN	CBCT	Prostate	DSC; Other; F1 Score	52
71	Bürkle et al. [129]	2024	CNN	CT	Prostate	DSC; SECE; TT	188
72	Zhang et al. [130]	2021	GAN	CT	Pelvic area	DSC; HD; Other	Not specified
73	Pan et al. [39]	2022	TTVNet; CNN; Vision Transformer (ViT).	CT	Prostate	DSC; HD; MSD; ReMSD; Other; TPR; ACC	234
74	Gibbons et al. [65]	2023	Commercial Software	CT	Head and neck; thoracic; Pelvic area	DSC; HD; SECE	400
75	Ghezzi et al. [131]	2023	CNN	PET	Prostate	DSC	85
76	Hoque et al. [132]	2023	Commercial Software	CT	Prostate; Head and neck	DSC; HD; RVD	40
77	Leung et al. [133]	2024	Masked-Net (3D U-Net)	MRI	Cervical	DSC; HD; MSD	23
78	Huang et al. [88]	2021	U-Net	MRI	Prostate; Cervical	DSC; JC; ASSD; 95HD	87
79	Wen et al. [134]	2024	DeepLabV3+; U-Net++; 3D U-Net	CT	Prostate	DSC; TS; CET	197
80	Yamauchi et al. [66]	2024	Commercial Software	CT	Head and neck; Prostate; Breast	DSC; SECE	120
81	Pera et al. [135]	2023	Commercial Software	CT	Lung; Breast; Prostate	DSC; RANK	95
82	Claessens et al. [136]	2022	CNN; Logistic Regression; SVM; RF	CT	Prostate	DSC; 95HD; APL	48
83	Zabihollahy et al. [93]	2021	3D U-Net	MRI	Cervical	DSC; 95HD	283
84	Chen et al. [31]	2022	CNN	MRI	Prostate	DSC	26
85	Cardenas et al. [137]	2025	3D U-Net	CT	Prostate	DSC; PQM; DM	1001
86	de Hond et al. [138]	2023	U-Net; cGAN	CBCT; CT	Prostate	DSC; MSD; 95HD; Other; MAE	56

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ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
87	Duan et al. [43]	2022	Commercial Software	CT	Prostate	DSC; MSD; 95HD; Other	107
88	Walker et al. [89]	2022	Commercial Software	CT	Prostate; Head and neck	DSC; DAP; IoV; CET	433
89	Marschner et al. [139]	2022	DI2IN	CT	Prostate; Cervical	DSC; 95HD; JC; Other; TPR; SPC	339
90	Constantinou et al. [140]	2025	Commercial Software	CT	Head and neck; Prostate	HD; DSC; SECE; Other	Not specified
91	Pemmaraju et al. [141]	2024	U-Net	CT	Prostate	DSC	30
92	Polymeri et al. [68]	2024	3D U-Net	CT	Prostate	DSC; HD; MSD	1530
93	Nachbar et al. [44]	2024	3D U-Net	MRI	Prostate	DSC; 95HD; APL	252
94	Starke et al. [74]	2024	Commercial Software	CT	Prostate	DSC; 95HD; TS; DM	30
95	Nourzadeh et al. [142]	2022	Ensemble architecture	CT	Lung; Prostate	ACC; F1 Score; DSC	311
96	Lempart et al. [52]	2023	U-Net; SENet-154	CT	Prostate; Anal	DSC; MSD; 95HD; TPR; SPC; ACC; F1 Score	2224
97	Palazzo et al. [143]	2023	Commercial Software	CT	Prostate	DSC; HD; IoV; TS	20
98	Lei et al. [144]	2021	Mask CNN	Transrectal Ultrasound (TRUS)	Prostate	DSC; MSD; 95HD; VD; Other	83
99	Miura et al. [145]	2024	Commercial Software	CT	Prostate	DSC; 95HD	10
100	Savenije et al. [40]	2020	CNN	MRI	Prostate	DSC; MSD; 95HD	150
101	Wang et al. [146]	2019	FCN	CT	Prostate	DSC; ASSD; TPR; PPV	313
102	Balagopal et al. [147]	2018	2D U-Net; 3D U-Net; ResNeXt	CT	Prostate	DSC; HD; ASSD	136
103	Dong et al. [148]	2019	cGAN; DAU-Net	CT; MRI	Prostate	DSC; HD; MSD; RMSD; VD; Other; TPR; SPC	140
104	Ma et al. [73]	2019	U-Net	CT	Prostate	DSC; SAR; DVH	70
105	Wang et al. [149]	2021	BCnet	CT	Prostate	DSC; DAP; TPR; PPV	313
106	Nouranian et al. [91]	2016	K-SVD	Transrectal Ultrasound (TRUS)	Prostate	DSC; HD; MSD	590
107	Kazemifar et al. [86]	2018	U-Net	CT	Prostate	DSC; HD; TPR; PPV	85
108	Fu et al. [150]	2020	cGAN	CBCT; MRI	Prostate	DSC; MSD	100

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ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
109	Oktay et al. [151]	2020	3D U-Net	CT	Head and neck; Prostate	DSC; HD; IoV	761
110	Weston et al. [152]	2020	3D U-Net	CT	Diverse malignancies	DSC; JC; Other; TPR	80
111	Kajikawa et al. [153]	2019	Commercial Software	CT	Prostate	DSC; DVH	95
112	Ahangari et al. [154]	2021	U-Net	PET; MRI	Cervical	Other; F1 Score; DVH	34
113	Kearney et al. [87]	2019	DAB-CNN	CT	Prostate	DSC; MSD	120
114	Zhang et al. [155]	2020	3D U-Net	CT	Cervical	DSC; HD; JC	91
115	Shi et al. [80]	2021	3D U-Net	CT	Cervical	DSC; ASSD	462
116	Zabihollahy et al. [156]	2022	3D U-Net	MRI	Cervical	DSC; 95HD; VD	125
117	Ma et al. [84]	2022	VB-Net	CT	Cervical	DSC; MSD; HD	66
118	Jiang et al. [157]	2024	Encoder-Decoder architecture	CT	Cervical	DSC; HD	216
119	Jiang et al. [158]	2021	RefineNet	CT	Cervical	DSC; HD	200
120	Yoganathan et al. [159]	2022	ResNet-50; InceptionResNetV2	MRI	Cervical	DSC; HD	Not specified
121	Wang et al. [72]	2024	3D U-Net	CT	Cervical	DSC; ASSD; 95HD	90
122	Chang et al. [160]	2021	3D U-Net	CT	Cervical	DSC; 95HD	400
123	Groendahl et al. [161]	2022	2D U-Net	CT; PET; MRI	Anal	DSC; 95HD; ASSD	122
124	Huang et al. [83]	2023	M-Net	CT	Cervical	ASSD; 95HD; VD; Other	53
125	Wang et al. [37]	2024	3D U-Net	CT	Cervical	DSC; JC; HD; Other	17
126	Arjmandi et al. [162]	2024	Encoder-Decoder architecture	CT	Prostate	DSC; HD; Other	28
127	Outeiral et al. [163]	2023	nnU-Net	MRI	Cervical	DSC; MSD; 95HD; Other	195
128	Rouhi et al. [164]	2024	Ensemble modeling	MRI	Cervical	DSC; 95HD; ASSD	166
129	Wang et al. [165]	2023	CNN	CT	Cervical	DSC; 95HD; JC; Other	60
130	Zhu et al. [76]	2024	U-Net	CT	Cervical	DSC; 95HD; SECE	98
131	Nie et al. [71]	2022	CNN; Encoder-Decoder architecture	CT	Cervical	DSC; 95HD; SECE; TPR	203
132	Tian et al. [166]	2023	Hybrid architecture (U-Net)	CT	Cervical	DSC; HD	276
133	Ding et al. [33]	2022	3D V-Net	CT	Cervical	DSC; HD; JC; ASSD	130

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ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
134	Beekman et al. [167]	2022	CNN	CT; CBCT	Cervical	DSC; MSD	84
135	Ni et al. [53]	2024	nnU-Net	MRI	Cervical	DSC; 95HD; APL	136
136	Sartor et al. [168]	2020	FCN	CT	Cervical; Rectal	DSC; MSD	266
137	Pei et al. [169]	2024	SVM; RF; KNN; XG-Boost; DT	CT	Cervical	ROC-AUC; TPR; ACC; SPC; F1 Score; Other	117
138	Chen et al. [62]	2022	3D U-Net	CT	Cervical	DSC; MDA; 95HD; Other	127
139	Wang et al. [170]	2020	Encoder-Decoder architecture	CT	Cervical	DSC; HD	125
140	Lempart et al. [171]	2022	3D U-Net	CT	Anal	DSC; 95HD; MSD; SECE	158
141	Hu et al. [172]	2021	U-Net	CT	Cervical	DSC; 95HD; Other	70
142	Xu et al. [173]	2021	Hybrid architecture (U-Net)	CT	Prostate	DSC; ASSD	186
143	Duprez et al. [174]	2023	nnU-Net	CT	Cervical	DSC; HD; 95HD; MSD; PPV	100
144	Cao et al. [175]	2022	CNN	CT; MRI	Cervical	DSC; TPR; PPV; 95HD; ROC-AUC	65
145	Rigaud et al. [176]	2021	DeepLabV3+; 3D U-Net	CT	Cervical	DSC; MDA; HD; Other	247
146	Liao et al. [177]	2023	AbsegNet	CT	Diverse malignancies	DSC; 95HD; Other	544
147	Peng et al. [178]	2024	3D U-Net	CT	Cervical	DSC; 95HD; ASSD	130
148	Liu et al. [95]	2024	U-Net	CT	Cervical	DSC; 95HD; SECE	156
149	Court et al. [179]	2024	Commercial Software	CT	Diverse malignancies	DSC; HD; SECE; Other	300
150	Hampole et al. [180]	2024	U-Net	Transrectal Ultrasound (TRUS)	Prostate	DSC; TPR; PPV; VD; MSD; HD	144
151	Vagni et al. [181]	2024	GAN	MRI	Prostate	DSC; 95HD; Other	60
152	Vagni et al. [54]	2024	GAN	MRI	Prostate	DSC; 95HD; Other	76
153	Czipczer et al. [182]	2023	3D U-Net	MRI	Head and neck; Pelvic area	DSC; PPV; TPR; MSD	209
154	Hu et al. [183]	2023	Commercial Software	CT	Diverse malignancies	DSC; PPV; TPR; HD	28

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ID	Authors	Year	Method	Image type	Cancer	Metrics	Dataset size
155	Chang et al. [184]	2024	Commercial Software	CT; CBCT	Cervical	DSC; Other	125
156	Muya et al. [185]	2023	Commercial Software	CT	Cervical	DSC; Other	30
157	Douglas et al. [186]	2024	Commercial Software	CT	Cervical; Head and neck	DSC; Other	93
158	Gutiérrez-Ramos et al. [187]	2025	DeepLabv3+	CT	Prostate	DSC; HD	100
159	Huang et al. [188]	2024	U-Net	CT	Rectal	DSC; JC; HD	148
160	Ecker et al. [189]	2022	U-Net	MRI	Cervical	DSC; MDA; ACC	56
161	Wang et al. [190]	2020	FCN	CT	Prostate	DSC; ASSD	313
162	Mohammadi et al. [191]	2021	ResU-Net	CT	Cervical	DSC; HD; ASSD; VC; PPV	113
163	Ju et al. [32]	2020	Dense V-Net	CT	Cervical	DSC; HD; JC	190
164	Ju et al. [192]	2021	Dense V-Net	CT	Cervical	DSC; HD; JC; Other	133
165	Amjad et al. [193]	2022	DCNN	CT	Head and neck; Pelvic area	DSC; MDA; SECE; Other	Not specified
166	Chapman et al. [194]	2022	Commercial Software	CT; CBCT	Prostate	DSC; MSD; Other	Not specified
167	Nemoto et al. [195]	2020	2D U-Net	CT	Prostate	DSC; Other	556
168	Fan et al. [196]	2022	Encoder-decoder architecture	CT	Cervical	TPR; PPV; DSC; Other	53
169	Schreier et al. [197]	2020	DCNN	CT; CBCT	Breast; Pelvic area	DSC; Other	754
170	Heilemann et al. [61]	2022	U-Net; cGAN	CT	Prostate	DSC; HD; MSD; PPV; TPR	308
171	Sibolt et al. [198]	2021	CNN	CBCT	Pelvic area	DSC; MDA; Other	39
172	Liu et al. [199]	2020	U-Net	CT	Cervical	DSC; 95HD	105
173	Torheim et al. [94]	2017	Fisher's Discriminant Analysis (LDA); Quadratic Discriminant Analysis (QDA)	MRI	Cervical	DSC; ROC-AUC; TPR; SPC; Other	78

Note: Metric abbreviations used in Table S1 (e.g., DSC, HD, ASSD, APL, SECE) are defined in Supplementary Table S4.

S2. Methods and architectures: detailed classification

Table S2 provides a comprehensive classification of the segmentation methods and architectures identified in the reviewed studies. The reported counts correspond to the number of studies in which each method was mentioned. Since multiple methods may be used within a single study, categories are not mutually exclusive.

Table S2: Detailed classification of segmentation methods and architectures reported in the reviewed studies. Counts indicate the number of studies in which each method was mentioned; categories are not mutually exclusive.

Category	Description	Methods included
Fully convolutional networks (FCNs)	Encoder–decoder architectures widely used for medical image segmentation.	U-Net family: U-Net (31), 3D U-Net (29), 2D U-Net (3), nnU-Net (5), U-Net3D (1), U-Net++ (1), ResU-Net (1), ResU-Net++ (1), ResU-Net3 (1), DCT-UNet (1), Masked-Net (1). Dense and hybrid variants: Dense V-Net (2), DpnU-Net (2), VB-Net (2), M-Net (1). Fully convolutional networks: FCN (4), FC-DenseNet (1), 3D V-Net (1).
Convolutional neural networks (CNNs)	Generic CNN-based segmentation models, including hybrid CNN-transformer approaches.	CNN-based models: CNN (19), DCNN (4), DAB-CNN (1), DeepLabV3+ (6), Mask R-CNN (1), ResNet-50 (2), ResNeXt (1). Hybrid transformer-based models: CNN-ViT (1).
Generative adversarial networks (GANs)	Adversarial learning methods used to generate realistic segmentations and improve generalization.	GAN-based methods: GAN (3), cGAN (7).
Machine learning (traditional and ensemble approaches)	Classical non-deep learning approaches, including statistical and ensemble methods.	Traditional ML algorithms: Random Forest (RF) (3), Support Vector Machine (SVM) (3), K-Nearest Neighbors (KNN) (1), Decision Trees (DT) (1), XGBoost (1), GBM (1). Linear discriminants: Fisher's Linear Discriminant Analysis (LDA) (1), Quadratic Discriminant Analysis (QDA) (1). Ensemble learning: Ensemble Modeling (1), Ensemble Machine Learning (EML) (1), Ensemble Architecture (1).
Atlas-based and probabilistic models	Methods based on anatomical priors and probabilistic or graph-based modeling.	Atlas-based methods: Atlas-Based (3). Probabilistic and graph-based models: Markov Random Field (MRF) (1), AbsegNet (1).
Specialized and hybrid architectures	Architectures that combine encoder–decoder structures with task-specific or hybrid components.	Encoder–decoder architectures: Encoder-Decoder Architecture (5), DAU-Net (1), RefineNet (2), RefineNetPlus3D (1). Other hybrid models: Hybrid Architecture (U-Net) (2), Context Encoder Network (CE-Net) (1). Multi-task learning models: Multi-Task Learning (1), TTVNet (1), PSA-Net (1), DI2IN (1).
Commercial software	Proprietary segmentation platforms used in clinical practice, often incorporating deep learning components.	U-Net-based: Limbus Contour (6), MVision (4), MIM Protege system (2), INTContour (2), DeepViewer (1). Atlas-based: Radiation Planning Assistant (3), STAPLE (1), ADMIRE (1). Hybrid models: Siemens AI-Rad Companion Organs RT (2), Siemens' Syngo.via Automatic Contouring System (1). GAN-based: DirectORGANS (1). CNN-based: Radformation (1). Not specified: DLCExpert (3), Varian Ethos system (2), OS-AIRIS (Project InnerEye) (2), TheraPanacea (1), RayStation (1), RapidPlan (1), AccuContour (1).

S3. Methodological quality, reproducibility, and validation indicators

Table S3: Distribution of methodological quality, validation, reproducibility, and reporting indicators across the reviewed studies ($n = 173$). The table summarizes the frequency of key indicators used to characterize evidential strength, transparency, and external applicability of the included studies.

Variable	Category	Count (n)	Percentage (%)
Validation and data-source characteristics			
Validation setting	Internal only	109	63.01
	Both	54	31.21
	External validation	6	3.47
	Not reported	4	2.31
Institution setting	Single-center	116	67.05
	Multi-center	55	31.79
	Not reported	2	1.16
Independent test set	Yes	163	94.22
	No	8	4.62
	Not reported	2	1.16
Scanner/protocol heterogeneity	Yes	92	53.18
	No	46	26.59
	Not reported	35	20.23
Reproducibility and transparency			
Public code availability	No	93	53.76
	Not reported	49	28.32
	Public	28	16.18
	Upon request	3	1.73
Public data availability	No	101	58.38
	Upon request	49	28.32
	Not reported	15	8.67
	Public	8	4.62
Annotator number reported	Yes	93	53.76
	No	74	42.77
	Not reported	6	3.47
Annotator expertise reported	Yes	153	88.44
	No	16	9.25
	Not reported	4	2.31
Consensus process reported	Yes	114	65.90
	No	53	30.64
	Not reported	6	3.47
Interobserver variability reported	No	115	66.47
	Yes	55	31.79
	Not reported	3	1.73
Clinical relevance, comparators, and optimization reporting			
Clinical or dosimetric evaluation	No	64	36.99
	Dosimetric	50	28.90
	Both	41	23.70

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Table S3: Distribution of methodological quality, validation, reproducibility, and reporting indicators across the reviewed studies ($n = 173$) (continued).

Variable	Category	Count (n)	Percentage (%)
Baseline comparison	Clinical	12	6.94
	Not reported	6	3.47
	Strong academic baseline	114	65.90
	No	35	20.23
	Commercial	14	8.09
Optimizer reported	Both	10	5.78
	Yes	104	60.12
	No	52	30.06
	Not reported	17	9.83
Tuning/hyperparameter strategy	Not reported	108	62.43
	No	34	19.65
	Yes	31	17.92
Evidence strength	Moderate evidential strength	95	54.91
	Lower evidential strength	54	31.21
	Higher evidential strength	24	13.87

Note: Percentages were calculated over the full set of included studies ($n = 173$). “Both” indicates studies reporting both internal and external validation, or both academic and commercial baseline comparison, depending on the variable.

S4. Definitions of evaluation metrics

Table S4 provides definitions of the evaluation metrics reported in the reviewed studies. The metrics are grouped according to their evaluation category (e.g., overlap-based, surface distance, classification, clinical and dosimetric, efficiency, and expert-based evaluation). This table clarifies the meaning of the abbreviations used in Table S1 and in the results presented throughout Section 3.

Table S4: Definitions of evaluation metrics used in the reviewed studies. Metrics are grouped according to their evaluation category.

Abbreviation	Metric	Description
Overlap-based metrics		
DSC	Dice Similarity Coefficient	Measures spatial overlap between predicted and reference segmentations. Values range from 0 (no overlap) to 1 (perfect agreement).
JC	Jaccard Coefficient	Intersection-over-union metric quantifying overlap between two segmentation volumes.
F1	F1 Score	Harmonic mean of precision and recall, commonly used in classification-based segmentation evaluation.
RVD	Relative Volume Difference	Quantifies the relative difference between predicted and ground-truth segmentation volumes.
VD	Volume Difference	Absolute difference between predicted and reference segmentation volumes.
Surface distance metrics		
HD	Hausdorff Distance	Maximum distance between the boundary points of the predicted and reference segmentation surfaces.

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Abbreviation	Metric	Description
95HD	95th Percentile Hausdorff Distance	Robust variant of Hausdorff Distance computed using the 95th percentile to reduce sensitivity to outliers.
MSD	Mean Surface Distance	Average distance between corresponding points on the surfaces of predicted and reference segmentations.
ASSD	Average Symmetric Surface Distance	Mean of the distances from each surface to the other, averaged symmetrically.
MDA	Mean Distance to Agreement	Mean distance between surfaces within a predefined tolerance region.
CLD	Centerline Distance	Distance between centerlines of predicted and reference anatomical structures.
CMS	Center of Mass Shift	Euclidean distance between centroids of predicted and reference segmentations.
APL	Added Path Length	Additional contour length required to correct the automated segmentation to match the ground truth.
Classification-based metrics		
ACC	Accuracy	Ratio of correctly classified pixels or voxels over the total number of samples.
TPR	True Positive Rate (Sensitivity / Recall)	Proportion of correctly identified positive pixels relative to the total positives in the reference segmentation.
SPC	Specificity	Proportion of correctly identified negative pixels relative to the total negatives in the reference segmentation.
PPV	Precision (Positive Predictive Value)	Proportion of predicted positive pixels that are correct.
FPR	False Positive Rate	Fraction of negative samples incorrectly classified as positive.
ROC-AUC	Area Under the ROC Curve	Measures classification performance across different decision thresholds.
Image similarity and regression metrics		
MAE	Mean Absolute Error	Average absolute difference between predicted and reference values.
MSE	Mean Squared Error	Mean of squared differences between predicted and reference values.
RMSE	Root Mean Square Error	Square root of the mean squared error.
SSIM	Structural Similarity Index Measure	Perceptual metric measuring structural similarity between images.
PSNR	Peak Signal-to-Noise Ratio	Measures image reconstruction quality relative to noise.
Correlation metrics		
r	Pearson Correlation Coefficient	Measures linear correlation between predicted and reference values.
ρ	Spearman Correlation Coefficient	Measures monotonic rank correlation between two variables.
Clinical and dosimetric metrics		
DVH	Dose–Volume Histogram Parameters	Quantitative parameters derived from dose–volume histograms used to evaluate radiotherapy plans.
HI	Homogeneity Index	Measures uniformity of radiation dose distribution within the target volume.
GI	Gradient Index	Quantifies the dose fall-off outside the target volume.
PCI	Paddick Conformity Index	Measures conformity of the prescribed radiation dose to the target volume.
PQM	Plan Quality Metrics	Set of quantitative indicators evaluating radiotherapy treatment plan quality.

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Abbreviation	Metric	Description
DCM	Dose Coverage Metrics	Measures evaluating coverage of the target volume by the prescribed dose.
DIT	Dosimetric Impact on Treatment Plans	Measures assessing the influence of segmentation differences on treatment planning dosimetry.
Efficiency metrics		
TS	Time Savings	Reduction in contouring or editing time achieved by automated segmentation.
CET	Contour Editing Time	Time required by clinicians to correct automated contours.
CCT	Contour Correction Time	Time required to adjust automatically generated contours.
Expert-based evaluation metrics		
IoV	Interobserver Variability	Variability among contours generated by different expert annotators.
SECE	Subjective Evaluation by Clinical Experts	Qualitative evaluation of segmentation results by clinicians.
CES	Contour Evaluation Scores	Scores assigned by experts evaluating segmentation quality.
RANK	Qualitative Ranking Score	Ranking-based evaluation comparing segmentation results across methods.
TT	Turing Test	Evaluation where experts judge whether contours were generated by humans or algorithms.

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